

Fourier Transforms and Heat Diffusion: Analytic Derivation and Numerical Efficiency

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1 Introduction

This study was an exploration of the use of Fourier transforms to solve the heat diffusion equation. There were two parts to this study: the first part was to understand how Fourier transforms are used analytically to derive the heat equation, and the other is how we can employ the Fast Fourier Transform algorithm to more efficiently solve diffusion-based temperature systems given discrete data. Our study finds that in both analytic derivation and discrete prediction and modeling of data, finding the solution in Fourier space and inverting it back to the spatial domain is a useful and efficient method for solving linear PDEs such as the heat diffusion equation.

To do so, first, the derivation of the Fourier transform from the trigonometric form of the Fourier Series is presented. Once the Fourier transform is defined, we demonstrate some relationships between the spatial and frequency domains that are utilized to solve the heat diffusion equation. Finally, the closed-form solution of the heat equation is derived and presented as Green's function, which is a convolution between a Gaussian distribution and the initial condition.

To demonstrate the numerical efficiency of the Fourier transform, two algorithmic implementations of the heat diffusion equation were devised: one implementing the Green's function closed-form solution, which involved $O(N^2)$ direct convolution, and one that performs a Fast Fourier Transform to perform the convolution in Fourier space and then convert it back to the spatial domain. First, these algorithms were applied and compared using a synthetic, one-dimensional initial condition that satisfied periodic boundary conditions to verify that outputs were identical and the FFT algorithm's efficiency would become increasingly better as the input size N grew.

Further verifying this comparison and exploring the utility of modeling the heat diffusion

equation, both models were applied to data retrieved from temperature sensors attached to the Curiosity Rover on Mars [8]. While air temperature cycles by solar day cannot be perfectly described by diffusion alone (there is solar radiation from the sun), the heat diffusion equation can serve as an accurate model where α represents thermal diffusivity, or how well temperature gradients are preserved across each sol. Using each algorithm iteratively, we found an α for two pairs of solar days: one on a clear day and one during a dust storm. We found that on a clearer day, more temperature gradient is preserved and thus we have a lower diffusivity and a better model. We also found that for an iterative process such as α fitting and for large enough input ($N = 30,000$), the FFT algorithm becomes the only feasible option for solving the PDE.

2 Background

Partial Differential Equations (PDEs) describe a wide variety of physical systems. While many of them can be solved using Separation of Variables or can be discretized and modeled with Finite Element Methods, for most it is usually difficult to find a closed-form solution. In the early 19th century, Joseph Fourier invented the Fourier transform for this purpose: to analytically solve the heat diffusion equation [5]. He discovered that we can transform a problem to the frequency domain, and in doing so, convert the PDE into an easily solvable Ordinary Differential Equation. The intuition behind this is that any function is expressible in terms of trigonometric functions, which is known as the Fourier Series. Using Euler's identity, we can express this series on the complex plane where x , a real valued variable that is typically interpreted as space, transforms to ξ , which is typically interpreted as frequency [2]. Once we have a solution in the frequency space, we invert the Fourier transform to give us a solution in the spatial domain. Because of the many convenient properties that define the relationship between the spatial and Fourier domain, linear PDEs are easily translated and solved in Fourier space to find closed-form solutions [4]. While we can take the transform of non-linear PDEs, they not as easily simplified and Fourier space and therefore their Fourier transforms do not offer much help.

We can also numerically solve PDEs with the Fast Fourier Transform, an algorithm that takes advantage of the roots of unity property $\omega_{2N}^{2k} = \omega_N^k$ to compute Fourier Series coefficients and perform a Fourier Transform in $O(N \log N)$ time. This idea was discovered by Carl Friedrich Gauss, who was calculating the orbits of asteroids, later rediscovered by John Tukey, who was trying to speed up detection of nuclear tests from seismic data, and officially published in 1965 by Cooley and Tukey [7].

My motivation for this study was to demonstrate how the heat diffusion equation is de-

rived from Fourier Transforms and to show its numerical efficiency in solving it discretely. Fourier transforms have an enormous amount of utility, as they are used in audio and signal processing, quantum mechanics, machine learning, image processing, and other PDEs and physical systems such as the Navier–Stokes equation and Schrödinger–Poisson system. Their versatility comes from, among other interesting properties, the ability to turn convolutions into multiplication, which makes analytic solutions possible and numerical solutions faster. Further, the Fourier transform is deeply connected with Euler’s identity and the complex plane, and therefore proves its utility in the real world, despite many arguing against the “reality” of the imaginary plane. My hope in this study was that, in furthering understanding of what makes the Fourier transform useful and efficient, I would be learning something new about everything to which it is applied, while also tracing how and why Joseph Fourier invented the Fourier transform and series in the first place.

3 Methods

In this section, we will derive the Fourier Transform and use it to derive the heat equation. In Section 8.5 of the *Numerical Analysis* textbook by R.L. Burden, J.D. Faires, and A.M. Burden [1], functions are transformed using the a trigonometric, numerical approximation of the Fourier Series by computing coefficients a_k , b_k , and with Fast Fourier Transform algorithms, c_k . The relevant formulas are the following:

$$S_n(x) = \frac{a_0}{2} + a_n \cos(nx) + \sum_{k=1}^{n-1} (a_k \cos(kx) + b_k \sin(kx))$$

where

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) dx \quad \text{for } k = 0, \dots, n \quad \text{and}$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx) dx \quad \text{for } k = 1, \dots, n-1$$

In Section 8.6 on Fast Fourier Transforms [1], a different form of $S_n(x)$ is expressed using c_k , a coefficient that combines a_k and b_k via Euler’s identity:

$$c_k = \sum_{j=0}^{2m-1} y_j e^{ik\pi j/m} \quad \text{and} \quad S_m(x) = \frac{1}{m} \sum_{k=0}^{2m-1} c_k e^{ikx}$$

, where

$$c_k = a_k + ib_k$$

For trigonometric formulas, the limitation is that we are confined to the interval $[-\pi, \pi]$ and that for any finite n the interpolation is not perfectly accurate. This is why we need the Fourier transform, which gives us the transform of any continuous function over $\mathbb{R}$ on an infinitely long interval [2].

3.0.1 Deriving the Fourier and Inverse Fourier Transform from the Fourier Series

Let us begin with the Fourier Series in exponential form on an interval $[-L, L]$ instead of $[-\pi, \pi]$:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L} \quad \text{and} \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

Let $k_n = \pi n/L$ and let $F(k_n) = \int_{-L}^L f(x) e^{-ik_n x} dx$.

Then, $c_n = \frac{1}{2L} F(k_n)$ and

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{F(k_n)}{2L} e^{ik_n x}$$

Notice that $\Delta k_n = \pi/L \implies \frac{1}{2L} = \frac{\Delta k_n}{2\pi}$, so

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{F(k_n)}{2\pi} e^{ik_n x} \Delta k_n$$

Our Fourier series is now a Riemann sum, so if take the limit of the Fourier series as our interval becomes infinitely large, $\Delta k_n \rightarrow 0$, $L \rightarrow \infty$ gives

$$\lim_{\Delta k_n \rightarrow 0} \sum_{n=-\infty}^{\infty} \frac{F(k_n)}{2\pi} e^{ik_n x} \Delta k_n = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$$

, where

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

If we interpret $F(k)$ as our original signal in the space or time domain, taking the complex Fourier series $f(x)$ as the interval $L \rightarrow \infty$ gives us the Fourier transform [2]. To derive the heat diffusion equation, we will use a slightly different convention for the Fourier transform than this equation. Nonetheless, this derivation demonstrates how the Fourier transform is the complex Fourier series as it is applied to an infinite interval and a continuous space.

3.0.2 Properties of the Fourier Transform

We will use the following definition of the Fourier Transform with respect to x [9]:

$$\mathbf{F}\{f(x)\} = \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{i\xi x} dx$$

Notice some differences in notation using the following definition. $f(x)$, which originally represented our complex Fourier series, is now the original function in the spacial domain. Further, we are using ξ for frequency instead of x and we use boldface and capitalization to represent Fourier transform $\mathbf{F}$, which is different than $F(k)$, which we used to define our original signal. We also have removed the scalar $\frac{1}{2\pi}$ and flipped the sign in the exponent, which is equally common in the literature. Albeit many changes have been made, This version is just as valid so long as we make the right adjustments to the Inverse Fourier Transform:

$$\mathbf{F}^{-1}\{\hat{f}(\xi)\} = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{-i\xi x} d\xi$$

These two components of the definitions are frequently swapped depending on the application, but the idea is the same. Here are two properties of Fourier Transforms that are central to what makes them so useful.

Derivative Theorem: [9]

$$\mathbf{F}(f'(x)) = -i\xi \mathbf{F}(f(x))$$

Derivatives in the spatial domain are multiplication in the frequency domain. This makes

sense given Euler's identity and the fact that $(e^{cx})' = c e^{cx}$. We can prove this with integration by parts and an assumption that $f(x)$ is well behaved, but we also have to make a important assumption that $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$. This has immense implications for application. It implies that for any interval on which we perform a Fourier transform on signal $f(x)$, we must satisfy boundary conditions. This is a condition that many PDEs and most certainly heat-diffusion satisfies (there cannot be temperature distributed everywhere on an infinite interval).

$$\mathbf{F}(f'(x)) = \int_{-\infty}^{\infty} f'(x) e^{i\xi x} dx = (f(x) e^{-i\xi x}) \Big|_{-\infty}^{\infty} + i\xi \int_{-\infty}^{\infty} f(x) e^{i\xi x} dx$$

Assuming that $f(x)$ is well-behaved. That is, as $x \rightarrow \pm\infty$, we know $f(x) \rightarrow 0$. We have

$$(f(x) e^{-i\xi x}) \Big|_{-\infty}^{\infty} = 0$$

and we are left with

$$i\xi \int_{-\infty}^{\infty} f(x) e^{i\xi x} dx = i\xi \mathbf{F}(f(x))$$

as desired.

Convolution and Inverse Convolution Theorem: [9]

$$\mathbf{F}\{(f * g)(x)\} = \hat{f}(\xi) \hat{g}(\xi) \quad \text{and} \quad (f * g)(x) = \mathbf{F}^{-1}(\hat{f}(\xi) \hat{g}(\xi))$$

$$\text{where} \quad (f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy$$

This can be proven using the Fourier transform and convolution definitions:

$$\mathbf{F}((f * g)(x)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) g(x - y) dy e^{-i\xi x} dx = \int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} g(x - y) e^{-i\xi x} dx dy$$

Let $s = x - y$ for some fixed y , then we have //

$$\int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} g(s) e^{-i\xi(s+y)} ds dy = \int_{-\infty}^{\infty} f(y) e^{-i\xi y} dy \int_{-\infty}^{\infty} g(s) e^{-i\xi s} ds = \mathbf{F}(f(y))\mathbf{F}(g(s))$$

as desired. With these theorems, we can analytically derive the solution to the heat equation.

3.1 Deriving the Solution to the Heat Equation

3.1.1 Problem Setup

The heat diffusion equation, which models how heat conducts over a homogeneous, one-dimensional material, is given by [5]:

$$u_t - c^2 u_{xx} = 0, \quad u(x, 0) = f(x), \quad c > 0, x \in \mathbb{R}, t \in \mathbb{R}$$

All we need is a diffusivity constant c , and we can understand how temperature $u(x, t)$ changes with respect to time and space. We know intuitively that temperature spreads out with time, so our solution should reflect that.

3.1.2 General Strategy

Our general strategy will be to transform this equation to Fourier space using the Derivative Theorem, which will convert this linear PDE into an ODE solvable with an integrating factor [5]. We then use the Convolution Theorem and Fourier inversion to recover the solution in space-time.

3.1.3 Fourier Transform of the Heat Equation

Although we have a two-variable equation, all Fourier Transform properties apply exactly the same. We take the Fourier Transform with respect to x only:

$$\mathbf{F}\{u_t\} = \mathbf{F}\{c^2 u_{xx}\}, \quad \mathbf{F}(u(x, t)) = \hat{f}(\xi)$$

By the Derivative Theorem applied twice:

$$\mathbf{F}\{c^2 u_{xx}\} = c^2 \mathbf{F}\{u_{xx}\} = -i\xi \mathbf{F}\{u_x\} = -\xi^2 \hat{u}(\xi, t)$$

Since the Fourier Transform is with respect to x and not t , we get $\mathbf{F}\{u_t\} = \partial_t \hat{u}(\xi, t)$, giving us the ODE:

$$\partial_t \hat{u}(\xi, t) = -c^2 \xi^2 \hat{u}(\xi, t) \quad \mathbf{F}(u(x, t)) = \hat{f}(\xi)$$

3.1.4 ODE Solution

Rearranging:

$$\partial_t \hat{u} + c^2 \xi^2 \hat{u} = 0$$

Using the integrating factor $\mu(\xi, t) = e^{c^2 \xi^2 t}$:

$$\frac{\partial}{\partial t} \left(e^{c^2 \xi^2 t} \hat{u}(\xi, t) \right) = c^2 \xi^2 e^{c^2 \xi^2 t} \hat{u}(\xi, t) + \partial_t \hat{u}(\xi, t) \cdot e^{c^2 \xi^2 t} = 0$$

by the product rule. Integrating both sides with respect to t :

$$e^{c^2 \xi^2 t} \hat{u}(\xi, t) = c(\xi) \implies \hat{u}(\xi, t) = e^{-c^2 \xi^2 t} c(\xi)$$

At $t = 0$ we have $c(\xi) = \hat{u}(\xi, 0) = \hat{f}(\xi)$, the Fourier transform of our initial condition.

Now notice that the solution in the frequency domain is the product of two functions $\hat{f}(\xi)$ and $e^{-c^2 \xi^2 t}$. Because $\hat{u}(x, t)$ is the product of two functions, by our Inverse Convolution theorem, $u(x, t)$ must be the convolution of the inverse Fourier transform of these two functions $\hat{f}(\xi)$ and $e^{-c^2 \xi^2 t}$:

$$\hat{u}(\xi, t) = e^{-c^2 \xi^2 t} \hat{f}(\xi) = \mathbf{F} \left\{ \mathbf{F}^{-1} \left\{ e^{-c^2 \xi^2 t} \right\} * f(x) \right\} \implies u(x, t) = \mathbf{F}^{-1} \left\{ e^{-c^2 \xi^2 t} \right\} * f(x)$$

We just need to find $\mathbf{F}^{-1}\left\{e^{-c^2\xi^2t}\right\}$.

3.1.5 Inverse Transform of the Diffusion Kernel

$$\mathbf{F}^{-1}\left\{e^{-c^2\xi^2t}\right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-c^2\xi^2t - i\xi x} d\xi$$

Completing the square in the exponent, treating it as a function w.r.t ξ , we get:

$$-c^2\xi^2t - i\xi x = -c^2t\left(\xi + \frac{ix}{2c^2t}\right)^2 - \frac{x^2}{4c^2t}$$

which gives:

$$\frac{1}{2\pi} e^{-x^2/(4c^2t)} \int_{-\infty}^{\infty} e^{-c^2t\left(\xi + ix/(2c^2t)\right)^2} d\xi$$

Setting $y = \xi + \frac{ix}{2c^2t}$ (so $dy = d\xi$) then $z = cy\sqrt{t}$ (so $dz = c\sqrt{t} dy$):

$$= \frac{1}{2\pi} e^{-x^2/(4c^2t)} \cdot \frac{1}{c\sqrt{t}} \int_{-\infty}^{\infty} e^{-z^2} dz$$

The Gaussian Integral. Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. Then [6]:

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \left(-\frac{1}{2} e^{-r^2} \right) \Big|_0^{\infty} = \pi$$

so $I = \sqrt{\pi}$. Therefore:

$$\mathbf{F}^{-1}\left\{e^{-c^2\xi^2t}\right\} = \frac{1}{2c\sqrt{\pi t}} e^{-x^2/(4c^2t)}$$

3.1.6 Aside: Connection to Quantum Mechanics

The Gaussian probability distribution function is given by $\frac{1}{\sqrt{2\sigma^2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$.

Notice that we started with a Gaussian kernel, and the Inverse Fourier Transform gave us another Gaussian kernel. This is an important property of Fourier transforms with implications for Heisenberg's uncertainty principle in quantum mechanics. The idea is that in physics, momentum is the Fourier transform of position. This Gaussian property combined with the scaling property of Fourier transforms tells us that the Fourier transform of a narrow

Gaussian will give a wider Gaussian. In other words, the more we know about the position of a particle, the less we know about its momentum. In the context of Gaussian wave functions this can be expressed as $\sigma_x \sigma_\xi = \frac{\hbar}{2}$, where $\hbar$ is the reduced Planck's constant [9].

3.1.7 Solution

Plugging this in, the solution to the heat equation is [5]:

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} e^{-x^2/(4c^2 t)} * u(x, 0)$$

A Gaussian convolved with the initial temperature distribution. As t increases, $\frac{1}{2c\sqrt{\pi t}} e^{-x^2/(4c^2 t)}$ describes a wider Gaussian. This is known as the diffusion kernel. As time progresses, we convolve the initial distribution with a wider and wider Gaussian, spreading and smoothing the distribution more and more, which is exactly the intuition about how heat behaves.

If we apply our analytic result to numerical data, we will see that it is also more efficient to solve the heat equation in Fourier space than in the spatial domain. By the Convolution Theorem, $\mathbf{F}\{(f * g)(x)\} = \hat{f}(\xi) \hat{g}(\xi)$, convolution in the spatial domain is multiplication in the frequency domain.

4 Application and Results

In this section, we compare the efficiency of numerically solving the heat equation in Fourier space using Fast Fourier Transforms and in the spatial domain using Green's function:

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} e^{-x^2/(4c^2 t)} * u(x, 0)$$

The discrete version of the Green's function convolution in the spatial domain is:

$$u[i, t] = \frac{1}{2c\sqrt{\pi t}} \sum_{j=0}^{N-1} u[j, 0] e^{-(i-j)^2 \Delta x^2 / (4c^2 t)}$$

Since we traverse all N inputs for every i , the complexity of direct convolution is $O(N^2)$.

However, as shown in the analytical derivation, convolution is multiplication in Fourier space. Thus, we can use the Fast Fourier Transform, which is $O(N \log N)$ [7], to transform the data to Fourier space, compute the point-wise product $\hat{f}(\xi) e^{-c^2 \xi^2 t}$, and inverse transform the result back to the spatial domain, also $O(N \log N)$. Combined, this is $O(N \log N)$.

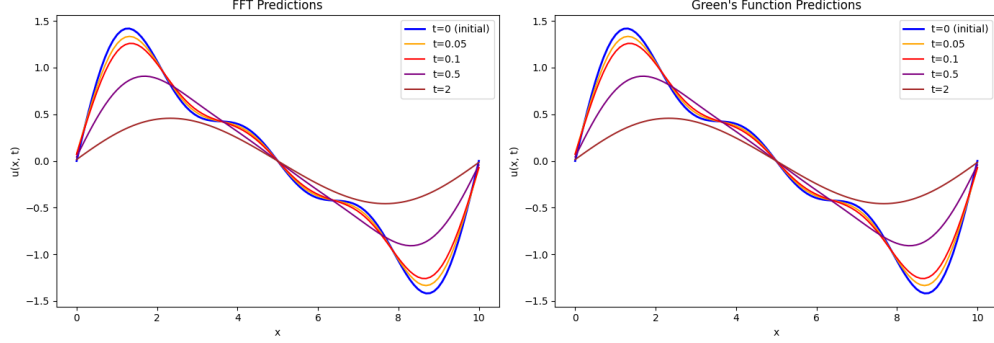


Figure 1: heat diffusion predictions at different time t overlaid onto initial condition $\alpha = 1, n = 100$

4.1 Benchmark on Synthetic Data

Both algorithms were implemented to demonstrate the difference in time complexity and test the heat diffusion equation's ability to fit real data. As a benchmark, a synthetic initial distribution was generated on an interval $L = 10$ from a truncated Fourier series

$$f(x) = \sin(5\pi x) + \sin(2\pi x/5) + \sin(3\pi x/5)$$

which is naturally periodic on $x \in [0, 10]$. From Figure 1, we can see that for all t , both methods produced equivalent results, which should be expected since they are analytically equivalent. However, for larger and larger input size N , the Fourier implementation became much faster, being 20,000 times faster for an input size of $N = 5,000$.

One important modification was required for the Green's function implementation. Instead of calculating the kernel directly in the spatial domain, we used an inverse transform of the FFT kernel. While Green's function and the Fourier implementation are theoretically the same, as mentioned in the derivation, this assumes an infinite interval. For a smaller, finite interval, and as time progresses, the Gaussian kernel's variance will be too large relative to the interval length L , so the kernel was modified. Even so, the direct convolution is still performed in the spatial domain, making the demonstration no less valid.

4.2 Application to REMS Data

To further test the efficiency difference and explore the heat equation's ability to fit real data, we retrieved data from the REMS meteorological package, which is designed to provide measurements of air and ground temperatures on Mars [8]. The data were received as telemetry from the Curiosity Rover and are formatted according to the REMS EDR SIS

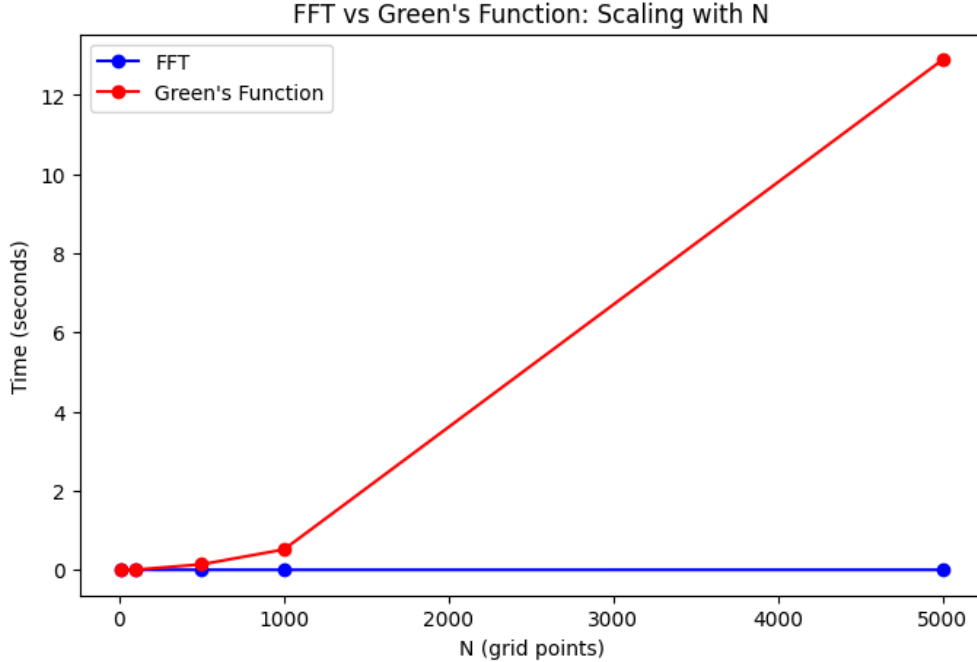


Figure 2: Algorithm times for different input sizes. $\alpha = 1, dt = 0.5$

specification. Sensor data are organized by Martian solar days (sols), and much of the data contain spikes from poor sensor conditions, so considerable effort was devoted to finding sols with accurate air temperature readings.

4.2.1 Data Preparation

After trial and error, we found that sensor data from the Boom 1 sensor was more accurate than the Boom 2 sensor and decided to drop data from the latter entirely. Data from two pairs of consecutive solar days — (2084, 2085) and (2358, 2359) — were used. As reported by the Mars Science Laboratory (MSL), for the former solar day pair a severe dust storm was occurring, and for the latter weather was very mild. All distributions were cleaned by shifting the mean to zero and slightly tilting the distributions to ensure periodicity. Sol 2358 had a small spike from sensor dysfunction which was removed by linear interpolation. Since temperature cycles on Mars are diurnal, each distribution was approximately sinusoidal and periodic, which as we saw in the benchmark is well-suited to heat diffusion modeling [8]. These distributions are shown in Figure 3.

4.2.2 α Fitting

Once the data were cleaned, both the direct convolution and FFT algorithms were used to find a diffusivity constant α that best fits the initial solar day's distribution onto the next

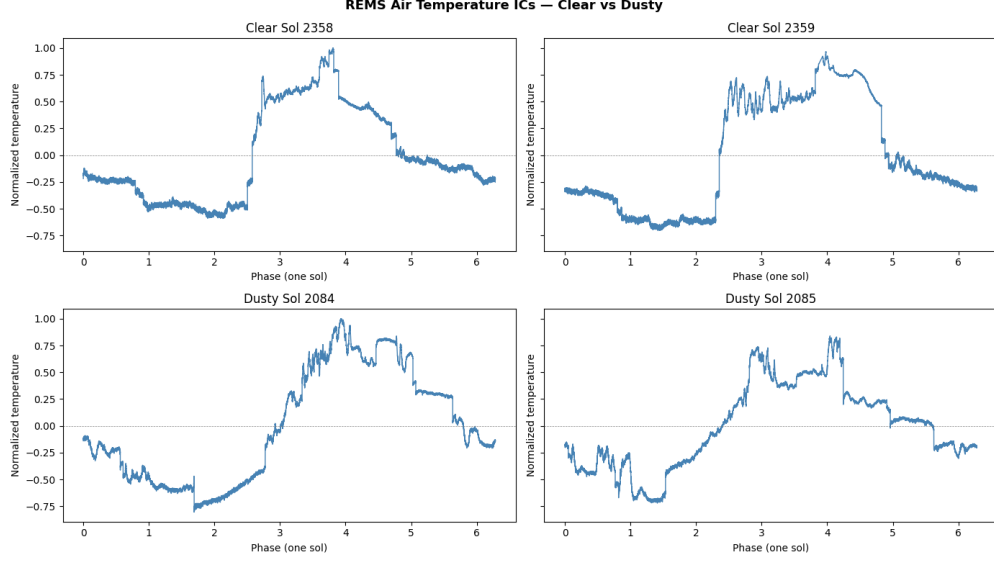


Figure 3: Cleaned air temperature distributions for each solar day. zero-meaned, normalized, and forced periodicity.

consecutive day. That is, for each initial sol N we sought an α such that, given $\Delta t = 1$ sol, the initial distribution diffused to resemble the next solar day’s distribution. To do so, we first normalized each pair of distributions by the maximum absolute value of their respective initial distributions to ensure equivalent scale. We also down sampled the distributions, which had an input size of $N = 30,000$, to $N = 1,024$, $2,048$, and $4,096$ for fitting α , as it would take days to perform direct convolution at the original size. We then iteratively tested α values in $[0.01, 10]$, computed the mean squared error, and returned the α with the least error. We did this with both methods and timed how long the iterative process lasted for each [4].

4.2.3 Results and Interpretation

As in the benchmark, the FFT algorithm performed much faster for both sol pairs. In Table 1, we see that FFT was approximately 9,162 times faster than direct convolution for pair (2358, 2359) and 25,116 times faster for pair (2084, 2085). Fitted α values were identical across methods, lending further credibility to each implementation. The low $\alpha = 0.1655$ for the clear-day sols means that relatively little change occurred in the temperature gradient across those days. In Figure 4, we can see that for the clear sol days, the predicted distribution fit the true data well, suggesting that diffusivity may be a strong model. However, for the dusty sols (2084, 2085), the mean squared error is much larger, and the diffusion equation did not fit the data as well, suggesting that the model is less appropriate during a dust storm.

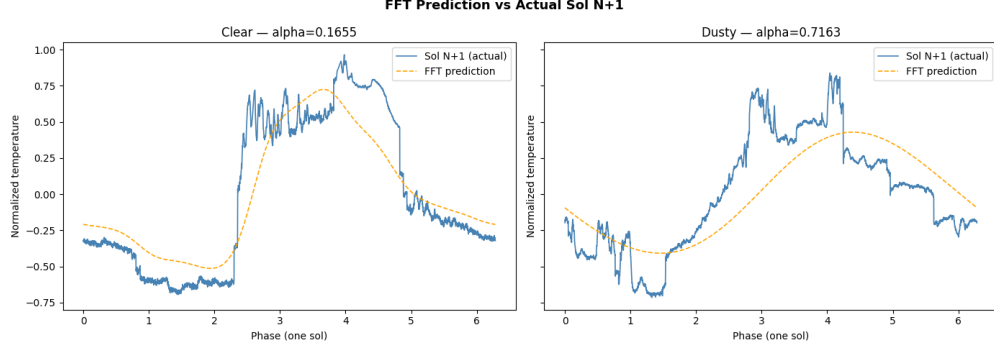


Figure 4: Initial distributions diffused with fitted α and overlaid onto distribution of next solar day.

One interpretive difficulty is that since we are not dealing with heat across space over time, our interpretation of α is ambiguous. At minimum, α is the rate at which the temperature profile is smoothed across consecutive days, and at best it represents the atmospheric thermal diffusivity. However, in truth we have not factored in other sources of heat such as solar radiation and convection, which limits the physical interpretation.

That said, α still has meaning when comparing the clear and dust storm solar days. The much higher and less well-fitting α during the dust storm suggests that temperature gradients changed more dramatically across those days. Because dust blocks solar radiation, a higher effective diffusivity would make sense since less radiation is being added to the system to counteract the diffusive smoothing.

However, in all cases the application of the heat diffusion equation to this system implies that the air temperature profile would continue to diffuse progressively across days, which is physically unrealistic since radiation is constantly being added to the system.

Table 1: Alpha fitting results for clear and dusty sol pairs.

Condition	Method	Fitted α	Time (s)	Speedup
Clear (2358 $\rightarrow$ 2359)	FFT	0.1655	0.010	9,162 $\times$
	Green's	0.1655	91.618	1.0
Dusty (2084 $\rightarrow$ 2085)	FFT	0.7166	0.002	25,116 $\times$
	Green's	0.7166	50.232	1.0

5 Final Comments

As demonstrated through the derivations and numerically with real temperature data, the Fourier transform is an efficient and useful tool for solving the heat equation. For finding the diffusivity constant for our Mars air temperature model, the original input size $N = 30,000$ was infeasible for direct convolution. Just as Tukey needed the FFT for seismic data that would have taken years to filter, the REMS data could only be fitted by our Fourier numerical solution of the heat equation in a reasonable amount of time. We have shown that the FFT, combined with the convolution theorem, converts convolution to multiplication in Fourier space, yielding an algorithm that makes problems of much higher input sizes tractable. The model produced from this data gave solid predictions for clear atmospheric conditions. Although not perfectly interpretable, the heat diffusion model’s accuracy suggests it as a good starting point for modeling diurnal air temperature cycles on Mars.

Some drawbacks include having to down sample and interpolate the REMS data, which may have altered the shape of the distributions and affected the fitted α values. With more time, better REMS data could have been found that did not require interpolation, and the full input size $N = 30,000$ could have been used to produce more accurate results. Additionally, having to modify the Green’s function kernel to use an inverse FFT does partially blur the distinction between the two methods, as it utilizes a Fourier transform at the kernel construction stage, though the dominant $O(N^2)$ convolution cost remains and the benchmark conclusion is unaffected.

There are numerous ways to build on this work. Analytically, deriving the complex Fourier series from the trigonometric Fourier series would add depth of understanding, as would formal proofs of the Derivative Theorem and Convolution theorem. On the applied side, modifying the heat equation to account for solar radiation could improve the fit for dusty sol pairs and make the physical interpretation of α more precise. To add more certainty to the results, fitting more sol pairs could help identify a universal or average diffusivity constant representative of atmospheric thermal diffusivity on Mars.

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